

POLY-VERIFICATION USER GUIDE

Welcome to PolyVerif

This user guide provides a detailed walkthrough for seamlessly conducting simulations with the Poly-Verification Suite utilizing the powerful PolyVerif framework. Before diving into the simulation process, make sure to refer to the comprehensive "InstallationGuide" for successful installation and configuration of the PolyVerif framework.

Configuration and Metric Defaults

Key configuration parameters are defined in the "config.ini" file, offering a glimpse into the default settings that govern the behaviour of the PolyVerif framework during simulations. These settings, ranging from detection thresholds to collision counts, play a vital role in the validation metrics that determine the success or failure of the simulation.

config.ini

```
[Autonomous_stack_config]
Detection_max_threshold = 90
Detection_min_threshold = 80
Control_collision_count = 0
Localize_max_threshold = 5
Localize_min_threshold = 1
Planner_goalpose_max_threshold = 5
Planner_goalpose_min_threshold = 2
```

Navigating the Simulation Landscape

1. Navigate to Poly_Suite Directory:

Go to the Poly_Suite directory located at adehome/Poly_Suite and execute the following commands:



```
shoukat@Shoukat: ~/adehome/Poly_Suite
shoukat@Shoukat:~$ cd adehome/Poly_Suite/
shoukat@Shoukat:~/adehome/Poly_Suite$ ./polyVerif
```

This command will initiate the PolyVerif framework for further configuration and execution

2. Select Validation Types for Detection

Four validation options await your exploration:

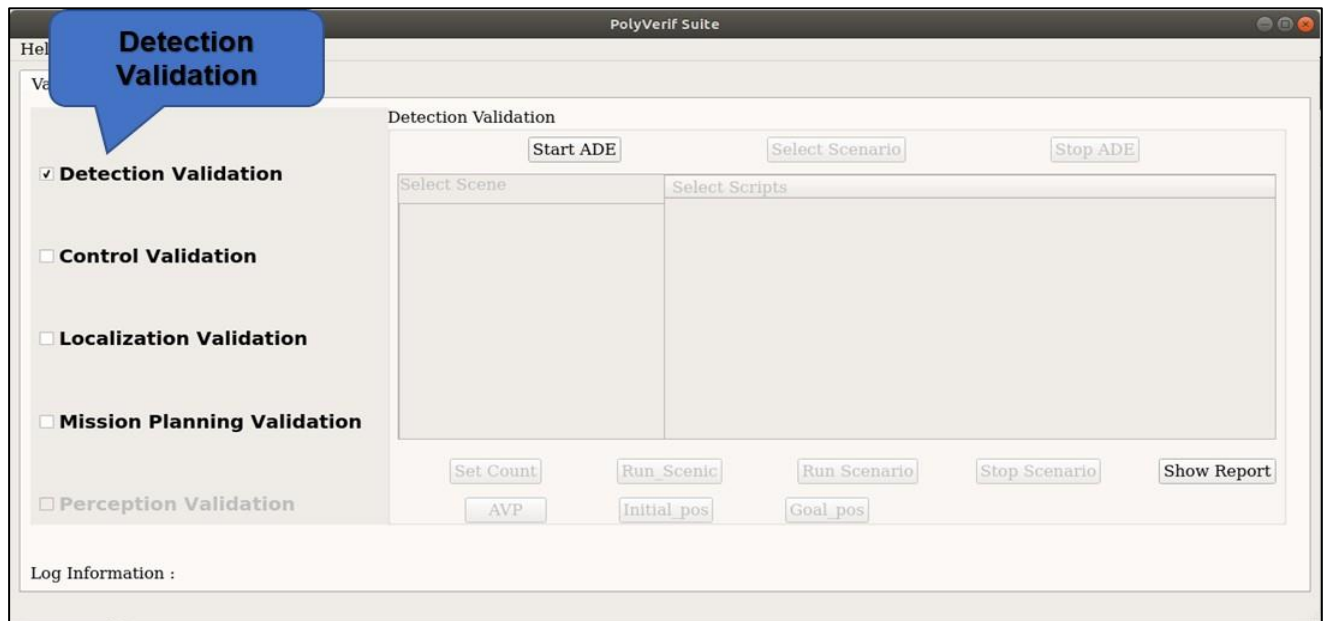
- **Detection Validation**
- **Control Validation**
- **Localization Validation**
- **Mission Planning Validation**



For the purpose of this guide, direct your focus towards Detection Validation and Mission Planning Validation.

Detection Validation:

Begin by clicking "Start ADE" to initialize modules, which includes Autoware, the Perception Stack, AWSIM simulator, Rviz, and Ros2.



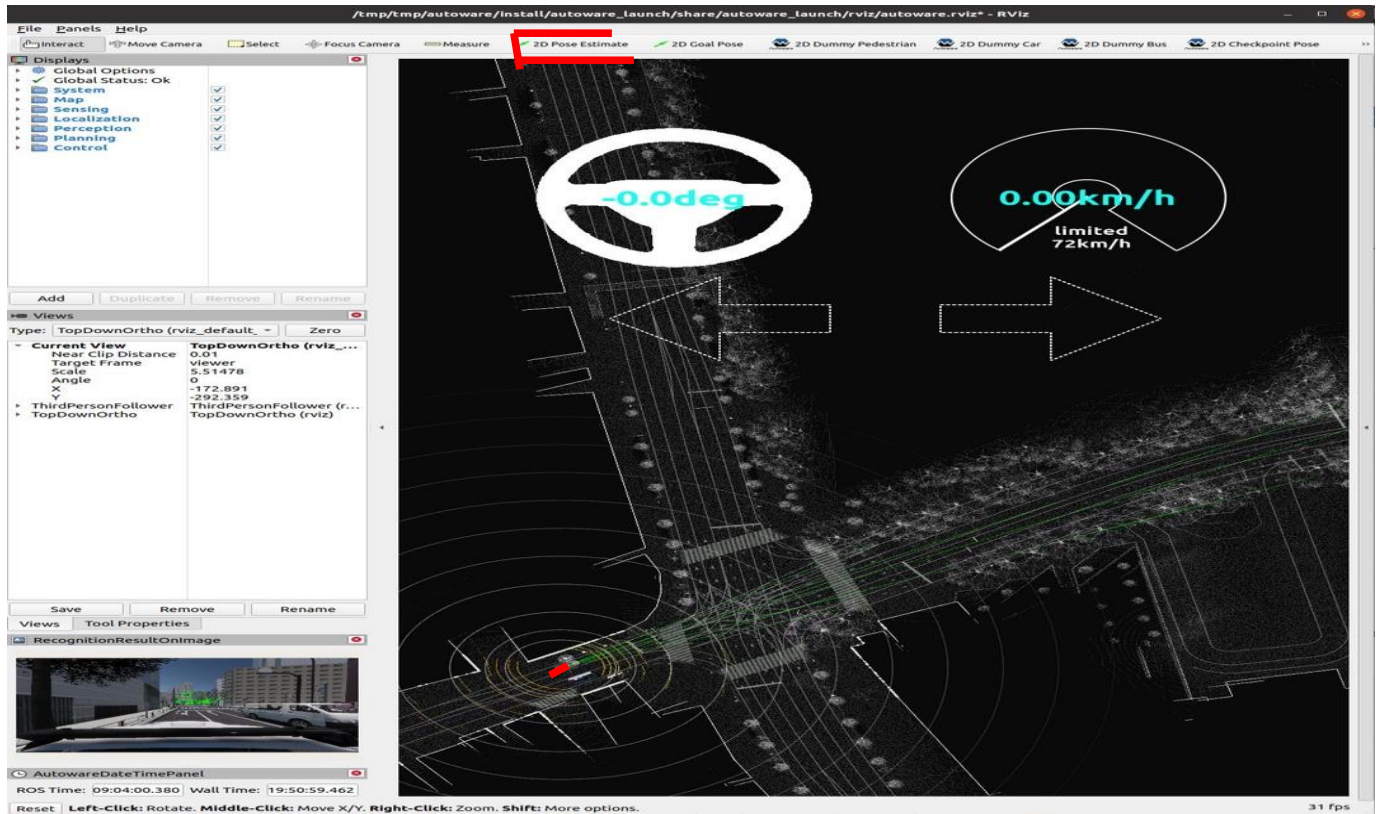
Allow some time for all necessary component get initialized in both simulator and rviz (map,ego,ego position,sensors etc).



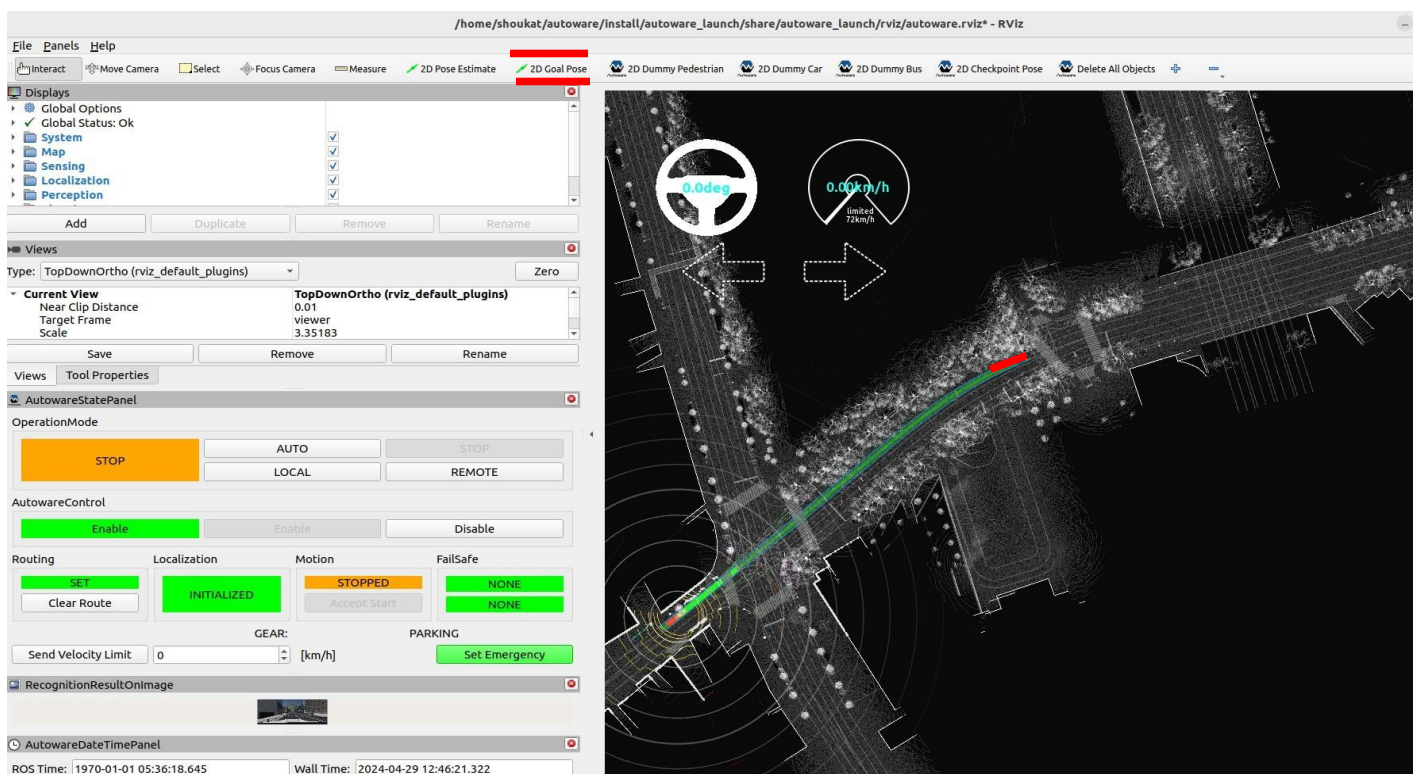
3.Set the initial and goal position

The Autoware will automatically set it's pose estimation as presented below.

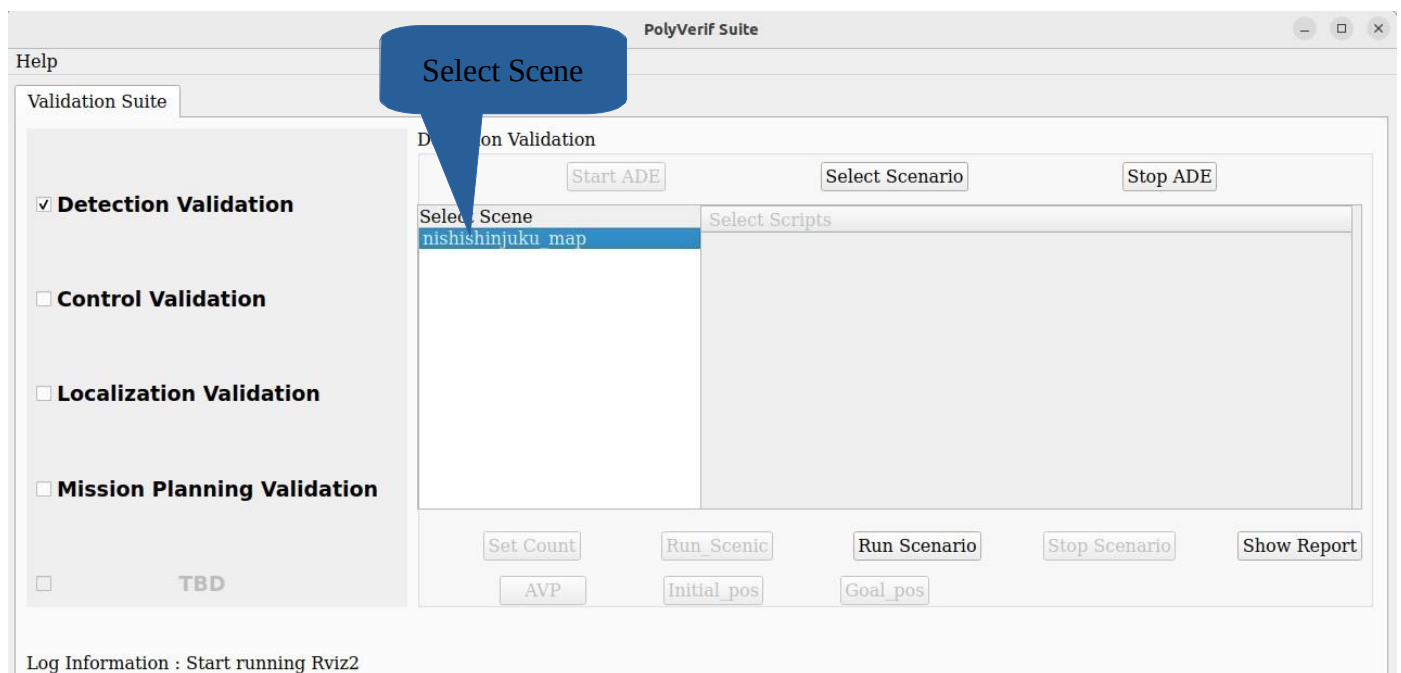
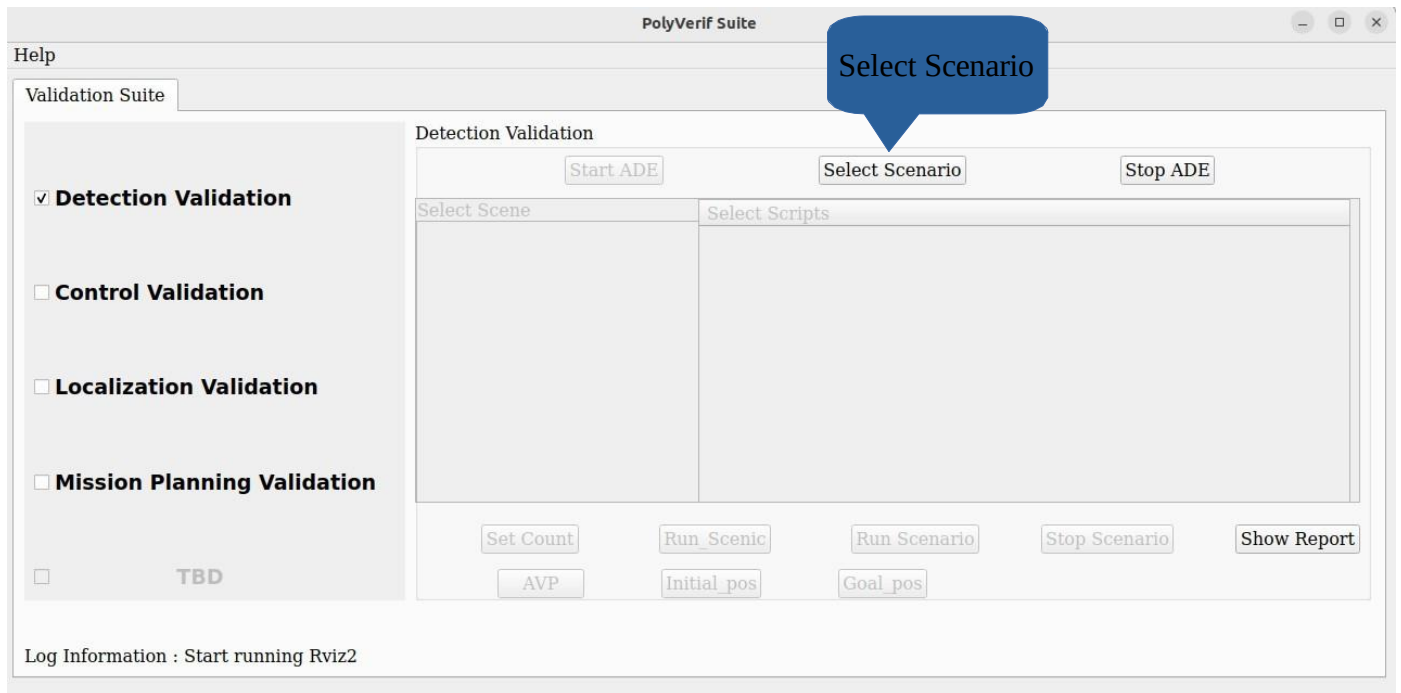
If the ego position not initializing properly then you can set ego position using **2D pose estimate** button and drag arrow on map.



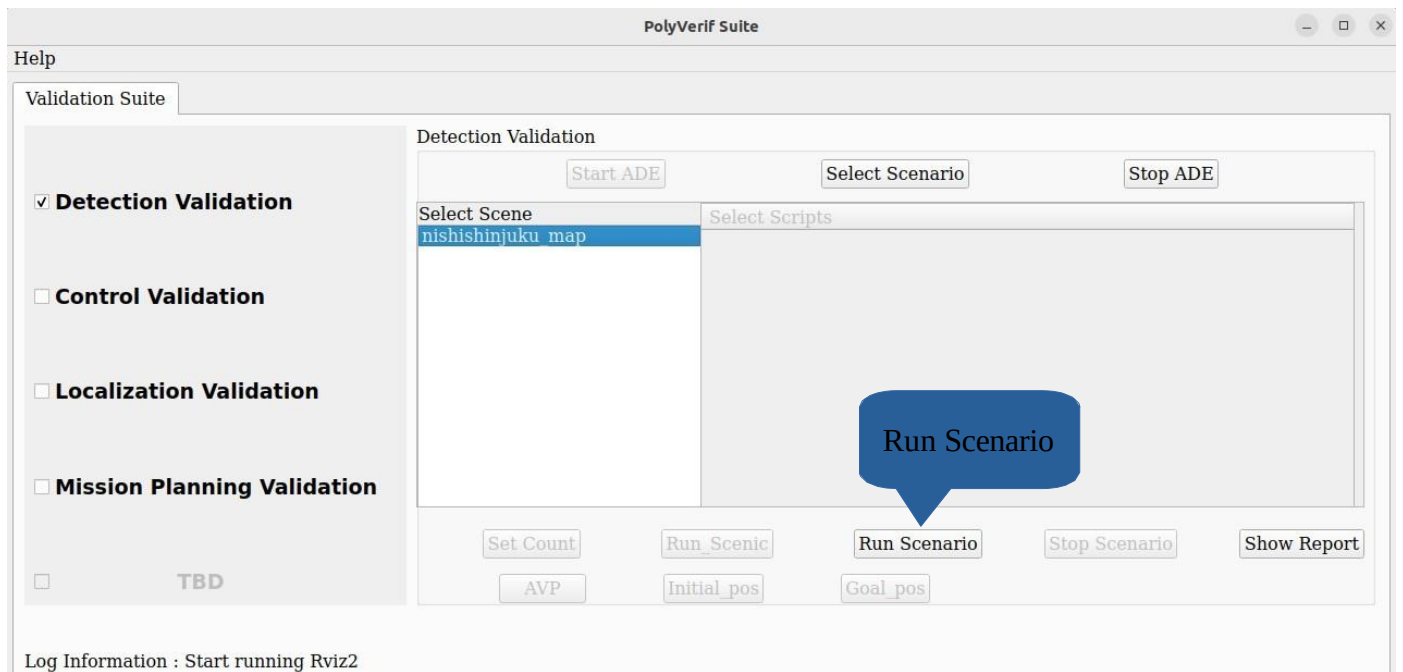
Set the navigation goal for the vehicle using **2D Goal Pose**.



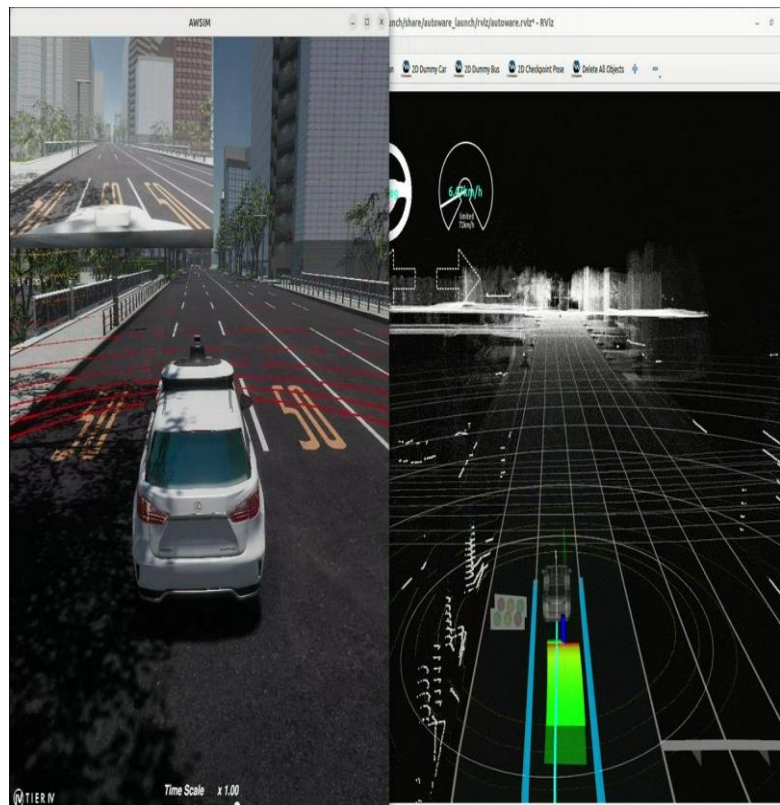
4. Select Scenario and Select Scene



5. Click on Run Scenario to Run Scenario:

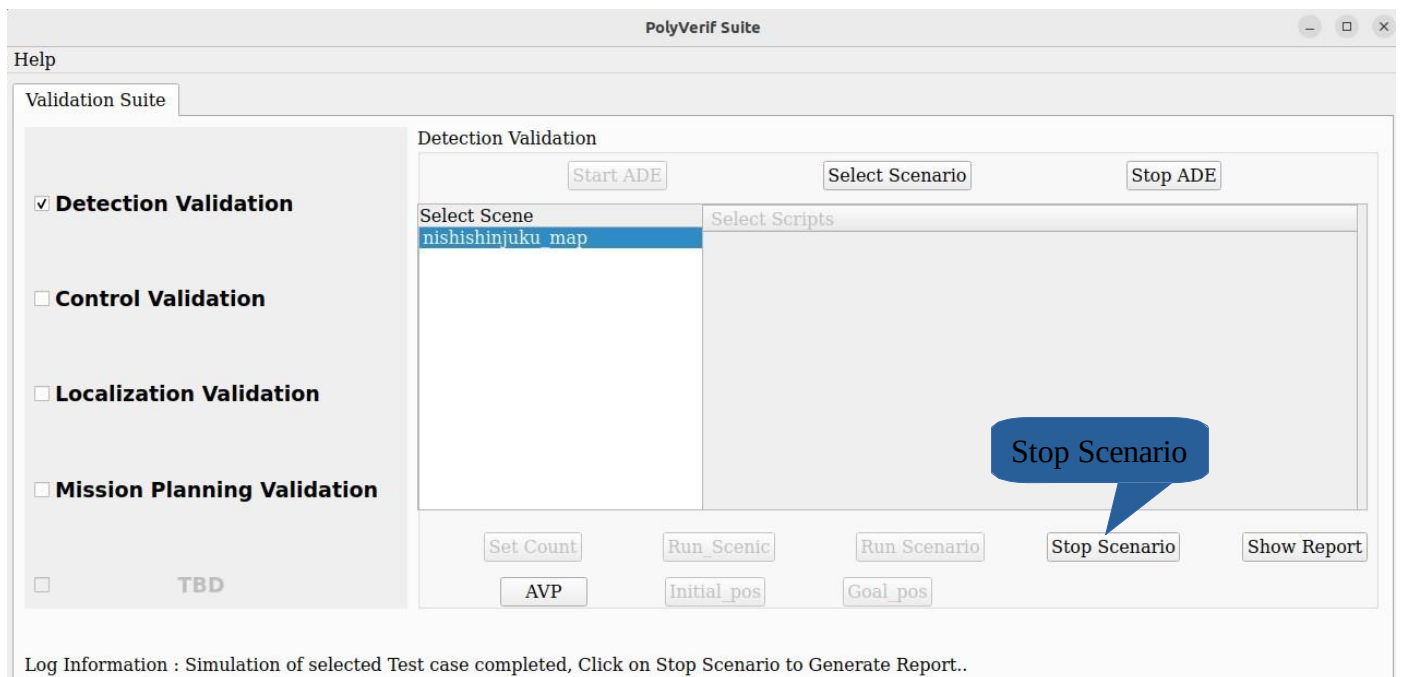


Let complete the scenario as shown as presented below.

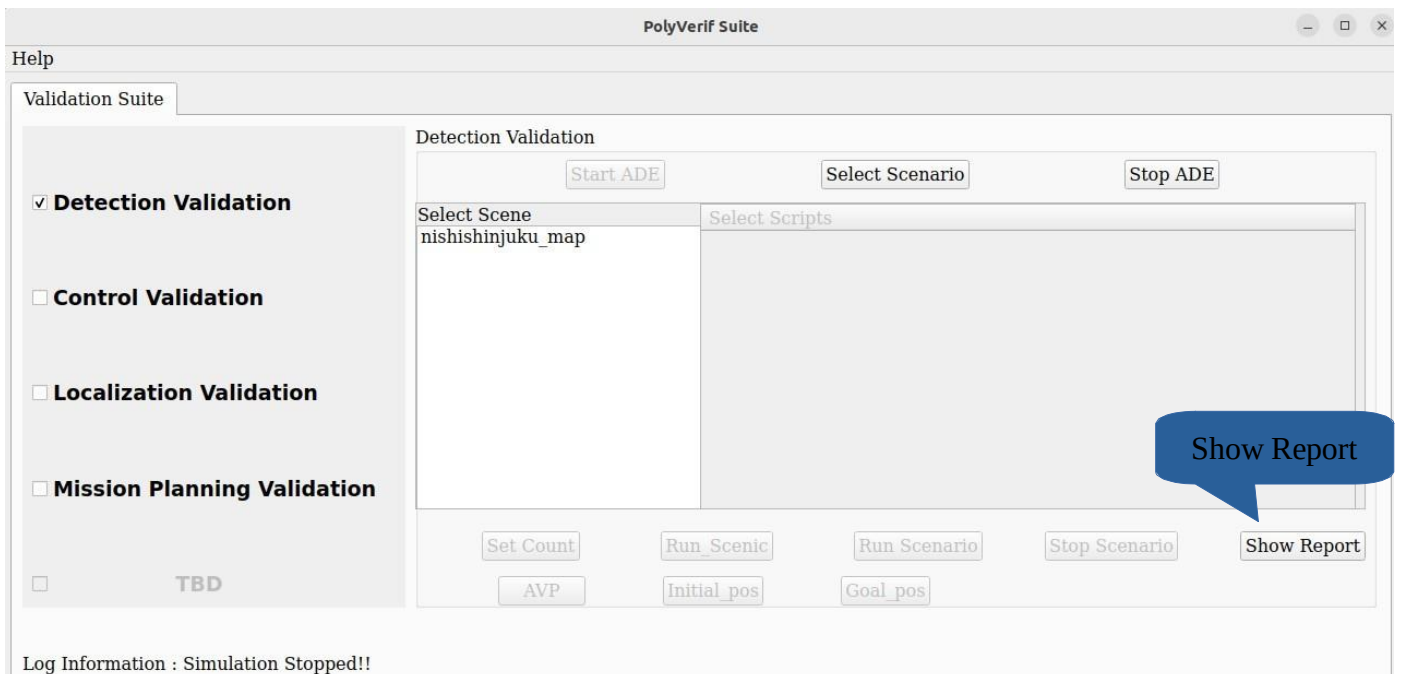


6.Simulation Management:

Utilize "Stop Scenario" to conclude the simulation.

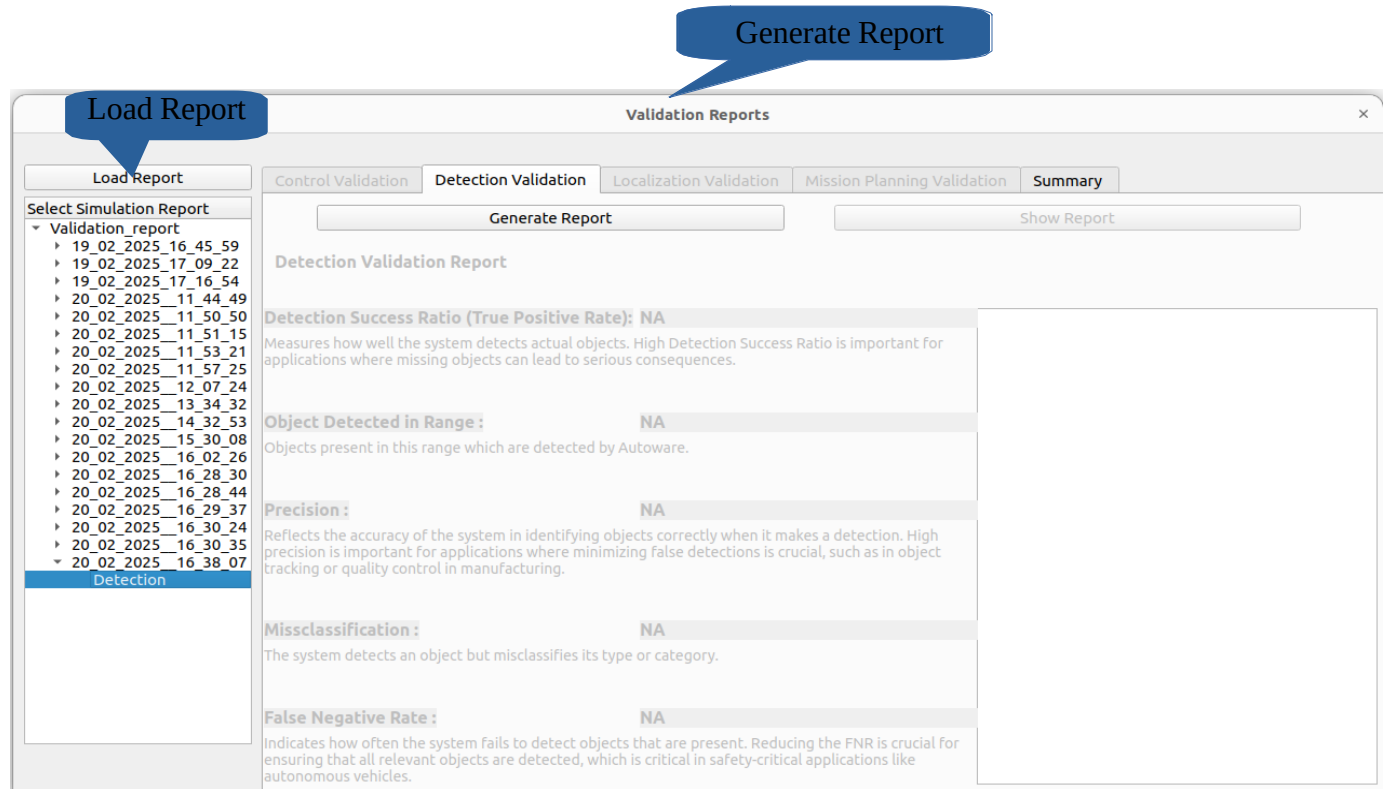


Access the detailed report through "Show Report."



7.Report Insights:

Explore simulation reports with the "Load Report" and "Generate Report" buttons.



Load Report

Generate Report

Validation Reports

Control Validation | **Detection Validation** | Localization Validation | Mission Planning Validation | Summary

Generate Report | Show Report

Detection Validation Report

Detection Success Ratio (True Positive Rate): NA
Measures how well the system detects actual objects. High Detection Success Ratio is important for applications where missing objects can lead to serious consequences.

Object Detected in Range : NA
Objects present in this range which are detected by Autoware.

Precision : NA
Reflects the accuracy of the system in identifying objects correctly when it makes a detection. High precision is important for applications where minimizing false detections is crucial, such as in object tracking or quality control in manufacturing.

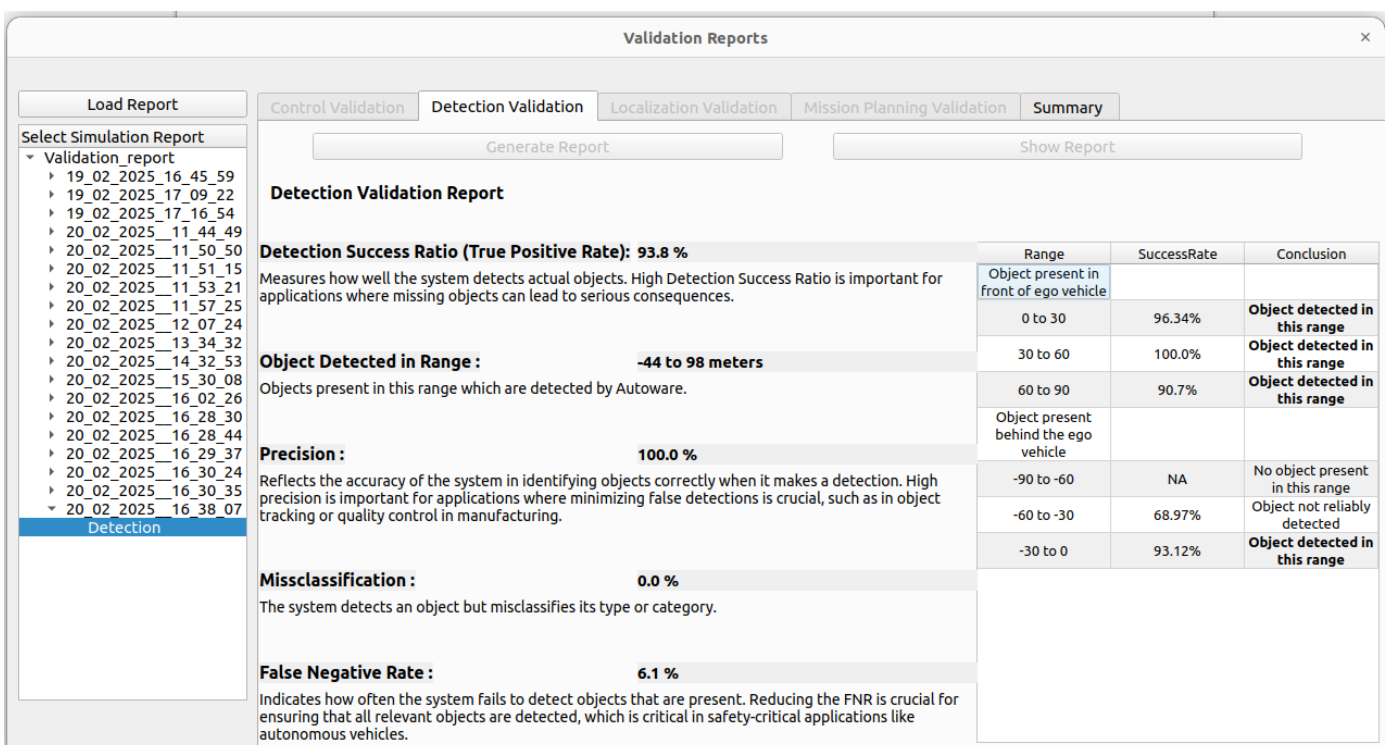
Missclassification : NA
The system detects an object but misclassifies its type or category.

False Negative Rate : NA
Indicates how often the system fails to detect objects that are present. Reducing the FNR is crucial for ensuring that all relevant objects are detected, which is critical in safety-critical applications like autonomous vehicles.

Note: It will take 2-3min to generate report

8.Report Management:

Explore simulation reports.



Load Report

Generate Report

Validation Reports

Control Validation | **Detection Validation** | Localization Validation | Mission Planning Validation | Summary

Generate Report | Show Report

Detection Validation Report

Detection Success Ratio (True Positive Rate): 93.8 %
Measures how well the system detects actual objects. High Detection Success Ratio is important for applications where missing objects can lead to serious consequences.

Object Detected in Range : -44 to 98 meters
Objects present in this range which are detected by Autoware.

Precision : 100.0 %
Reflects the accuracy of the system in identifying objects correctly when it makes a detection. High precision is important for applications where minimizing false detections is crucial, such as in object tracking or quality control in manufacturing.

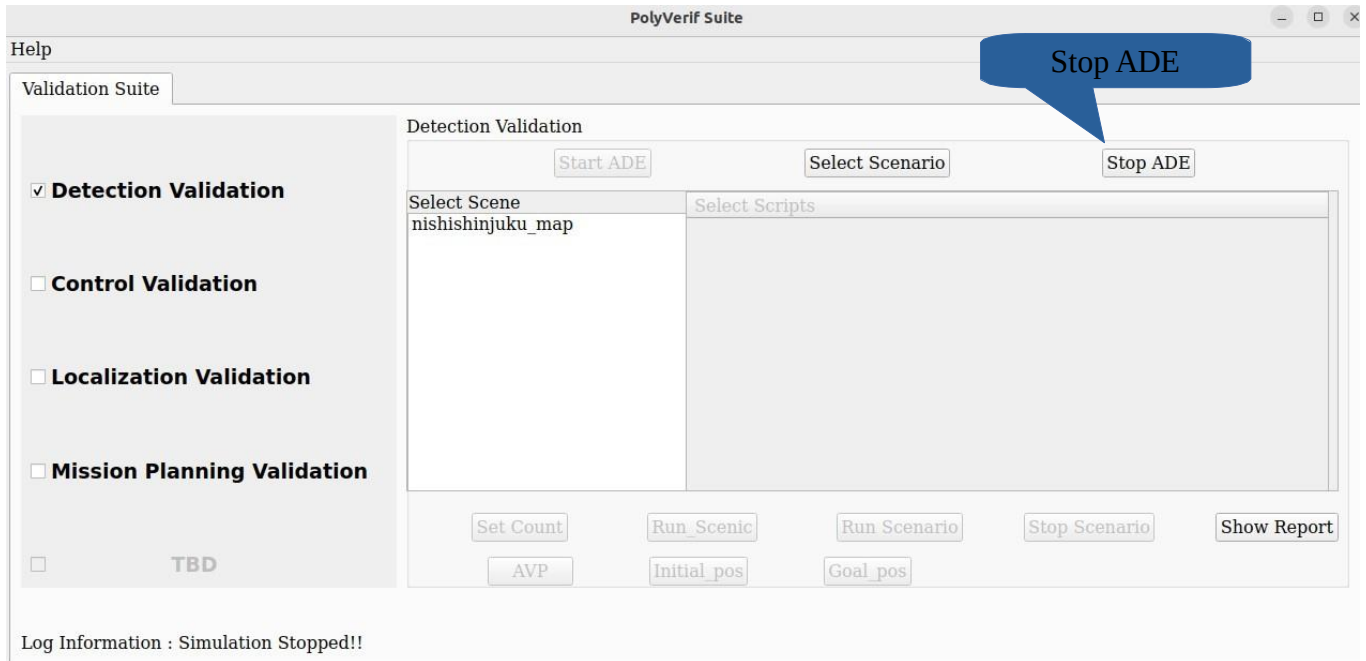
Missclassification : 0.0 %
The system detects an object but misclassifies its type or category.

False Negative Rate : 6.1 %
Indicates how often the system fails to detect objects that are present. Reducing the FNR is crucial for ensuring that all relevant objects are detected, which is critical in safety-critical applications like autonomous vehicles.

Range	SuccessRate	Conclusion
Object present in front of ego vehicle		
0 to 30	96.34%	Object detected in this range
30 to 60	100.0%	Object detected in this range
60 to 90	90.7%	Object detected in this range
Object present behind the ego vehicle		
-90 to -60	NA	No object present in this range
-60 to -30	68.97%	Object not reliably detected
-30 to 0	93.12%	Object detected in this range

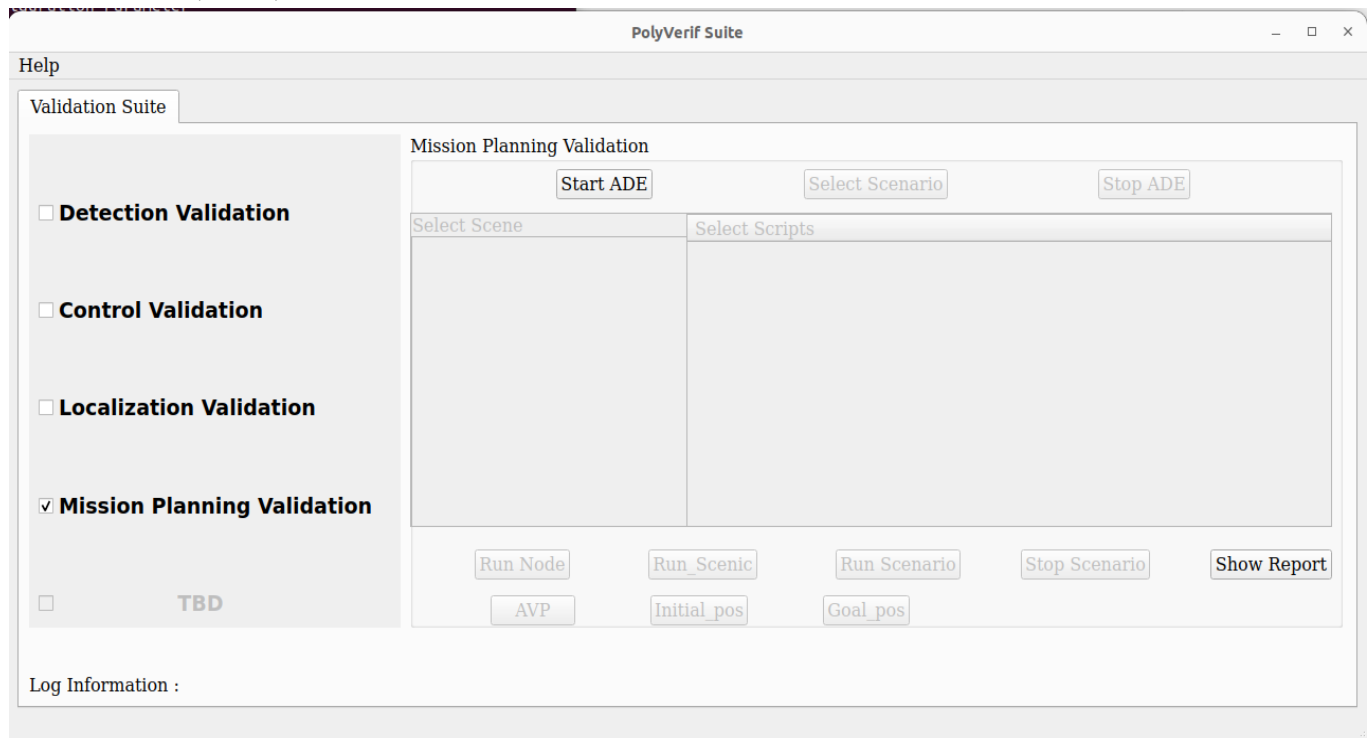
9. Stopping ADE:

After completing the scenario, click on "Stop ADE" to halt the PolyVerif framework and then close the terminal.



Mission Planning Validation:

Begin by clicking "Start ADE" to initialize modules, which includes Autoware, the Perception Stack, AWSIM simulator, Rviz, and Ros2.



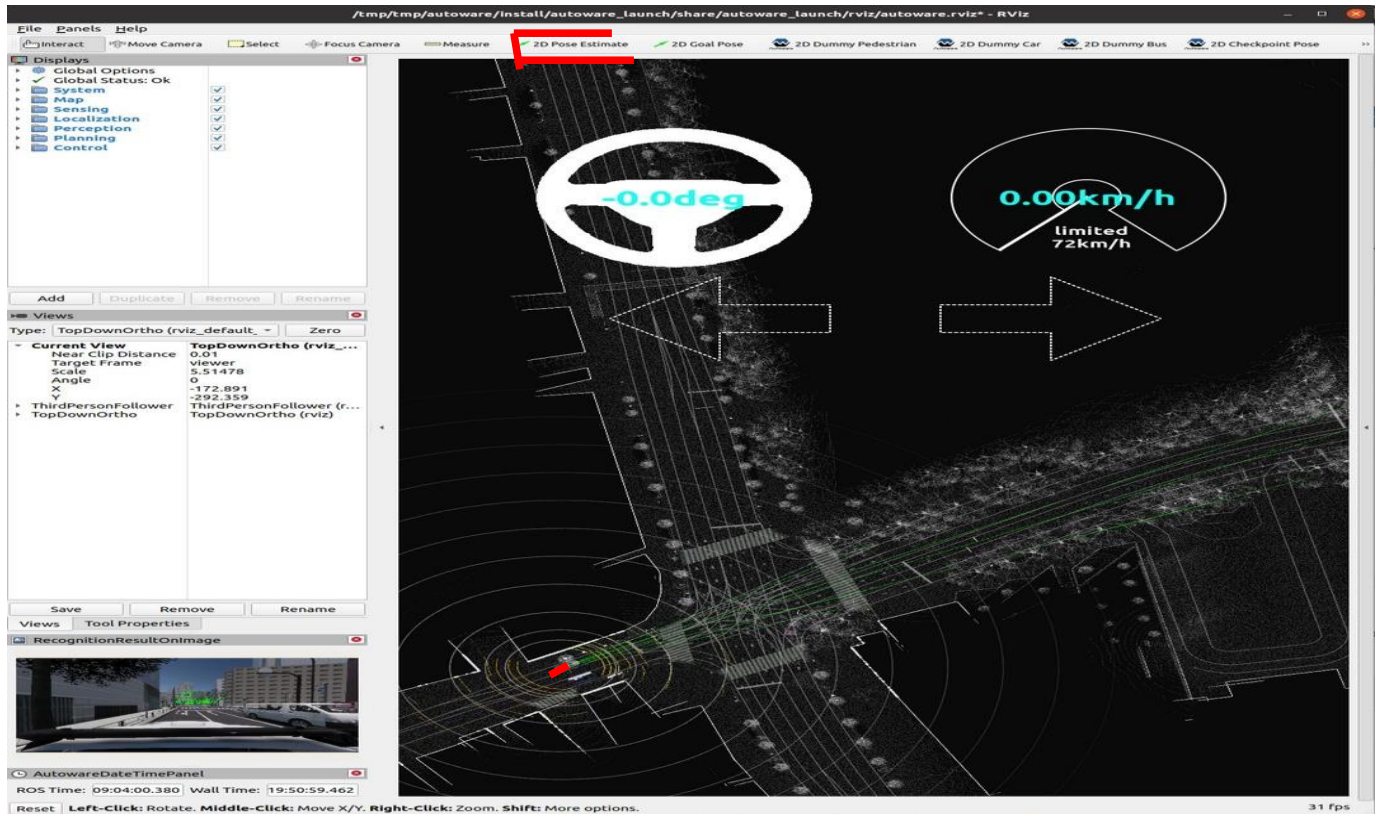
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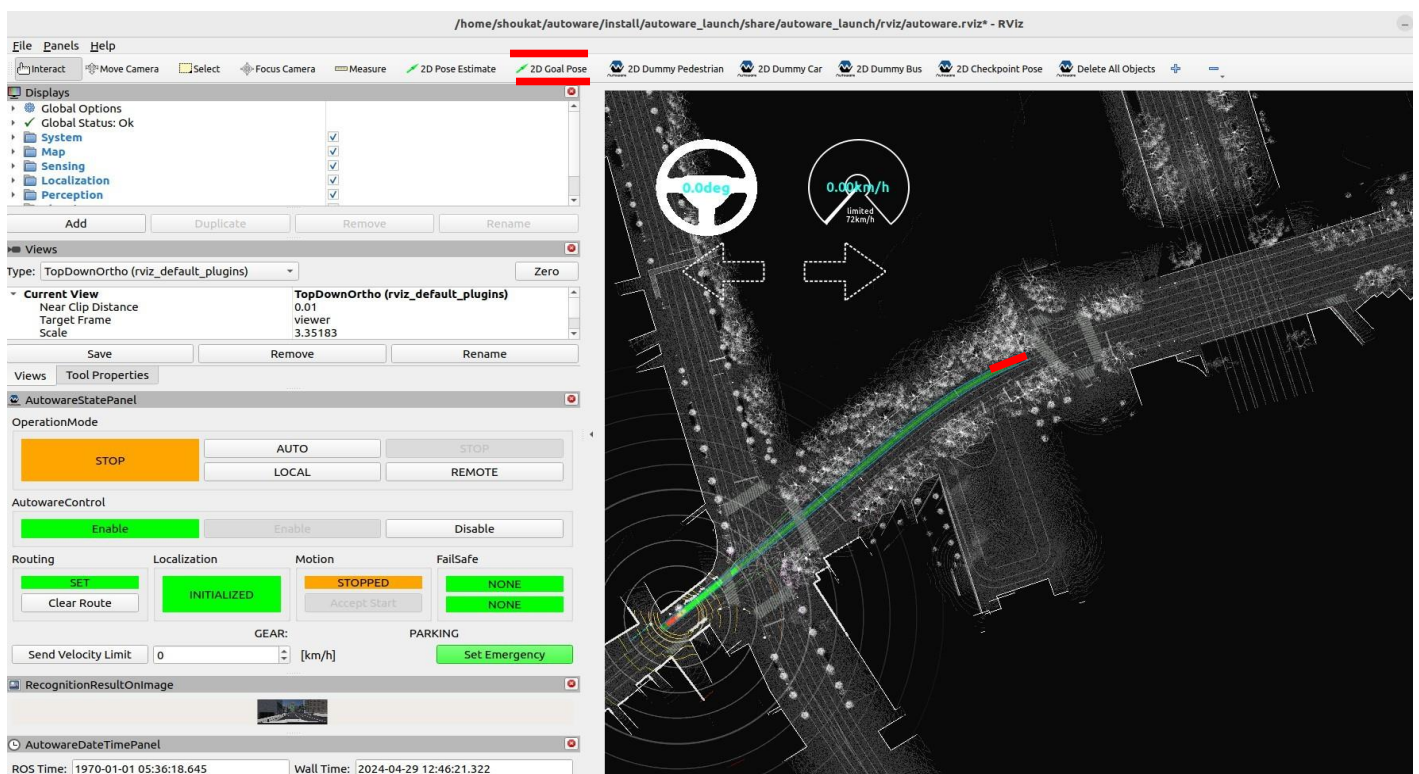
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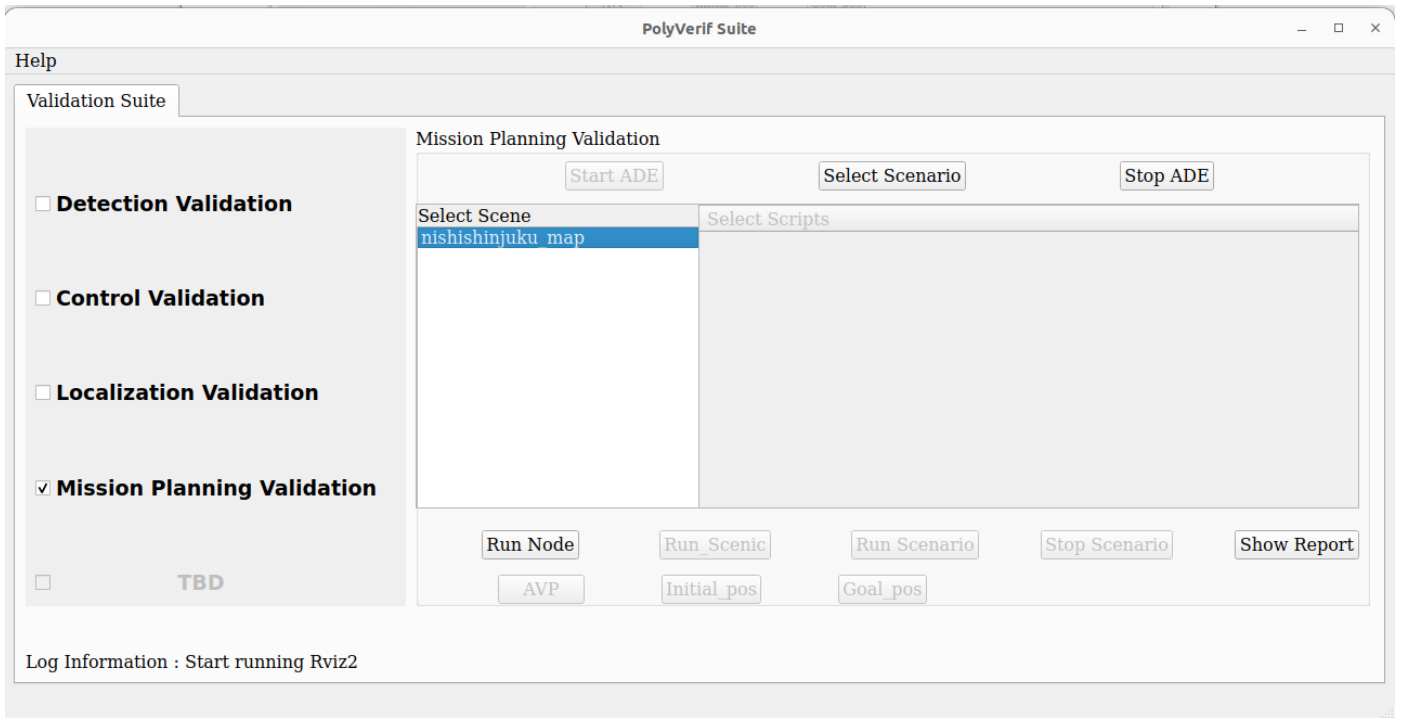
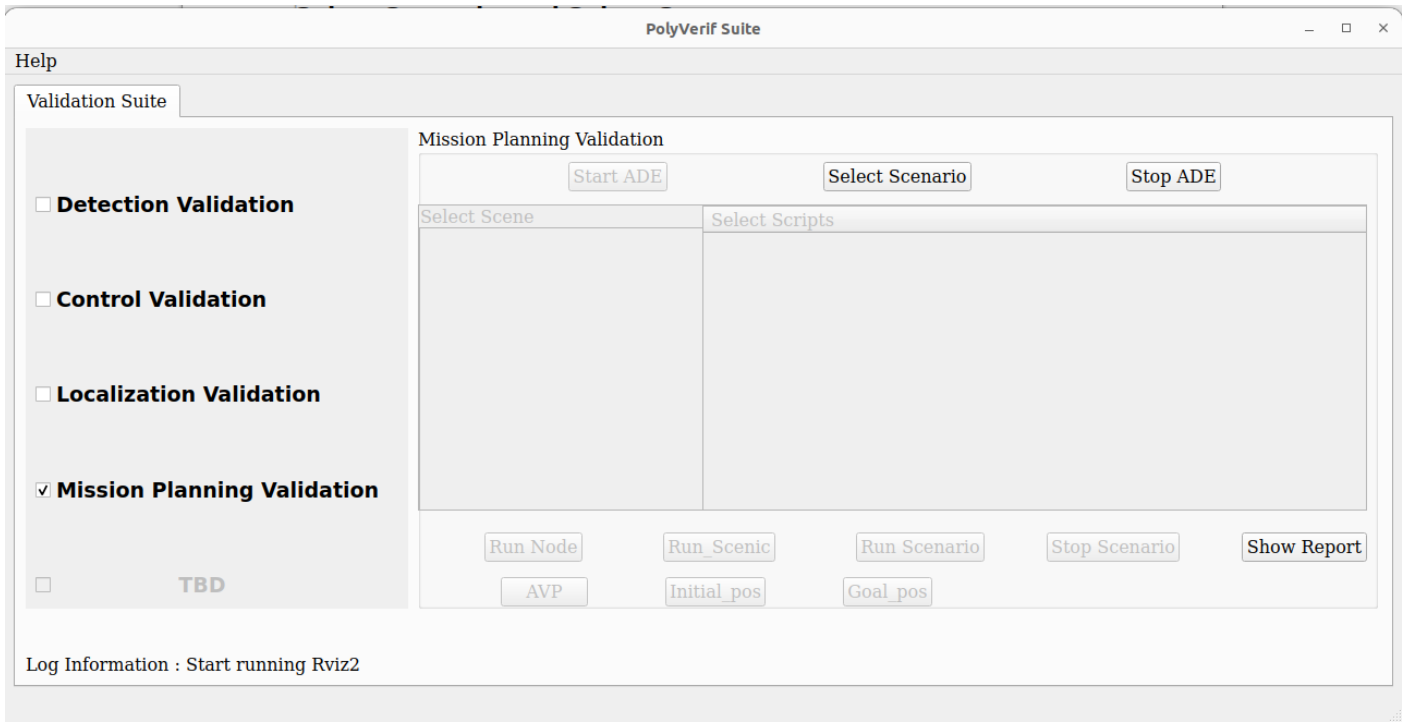
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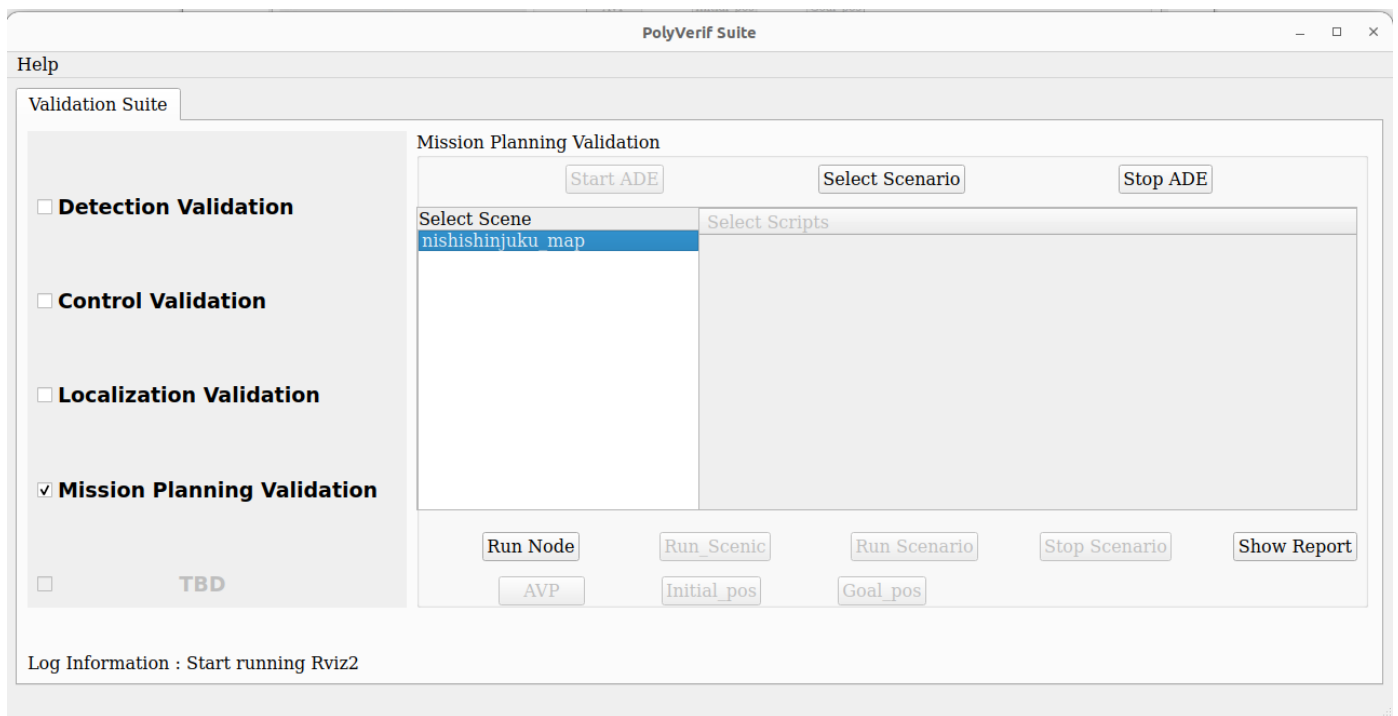
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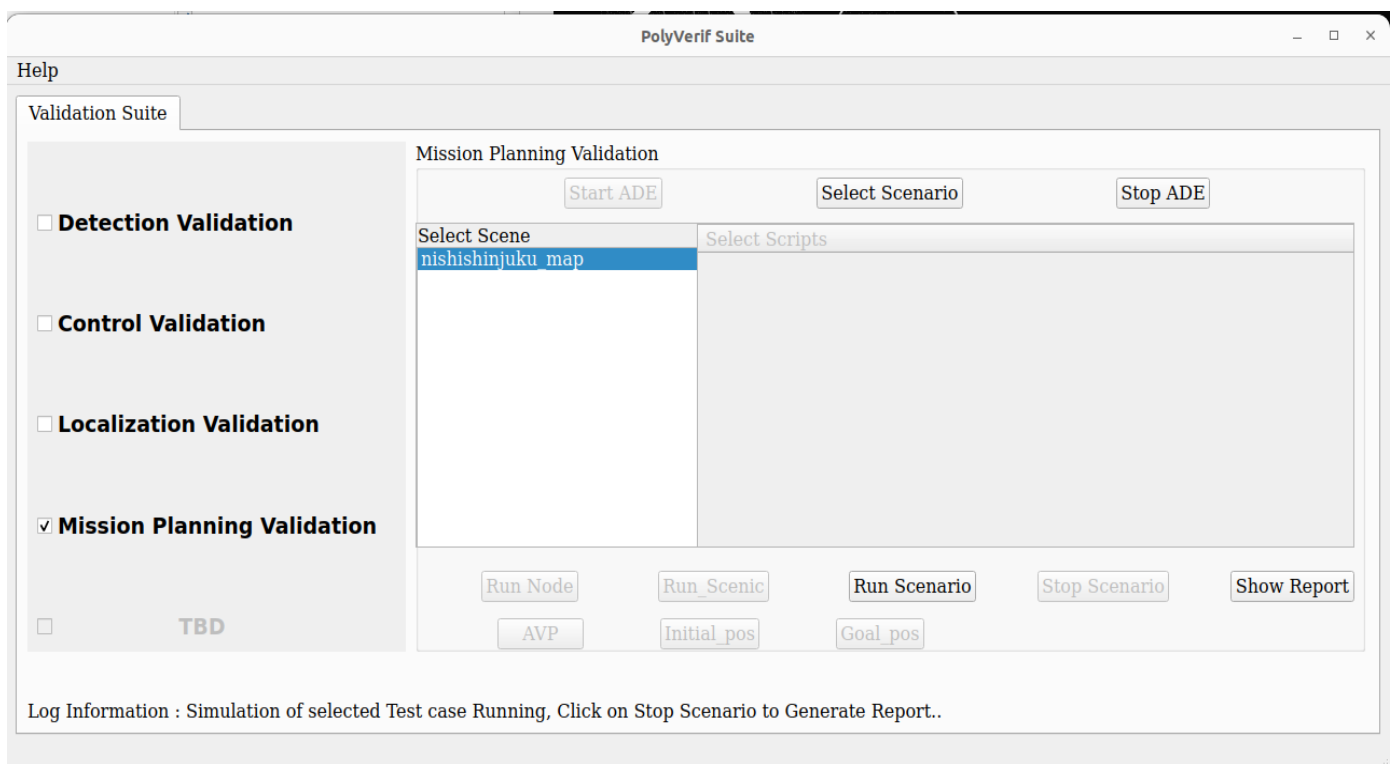
2. Select Scenario and Select Scene



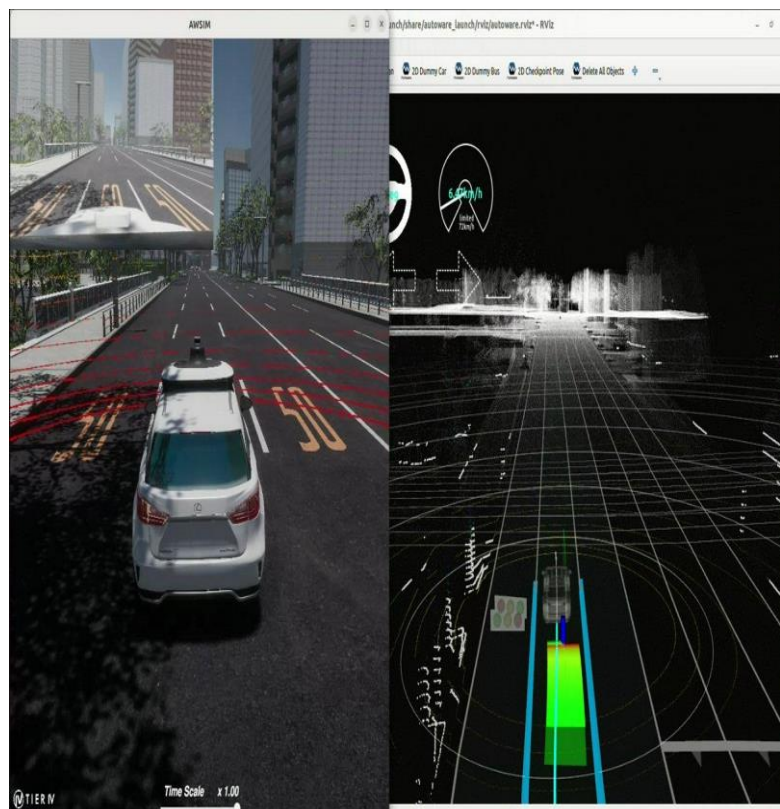
3.Click on Run Node to Run Mission Planning Node:



4.Click on Run Scenario to Run Scenario:

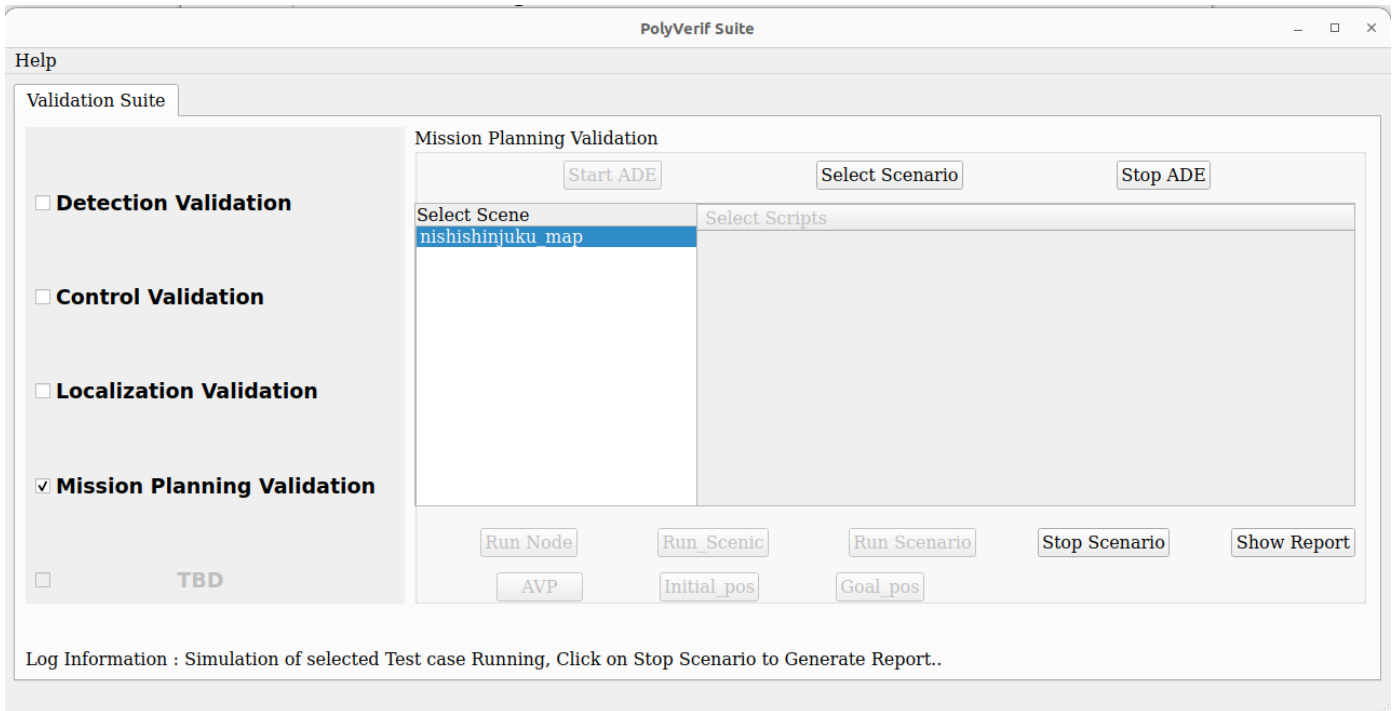


Let complete the scenario as shown as presented below.

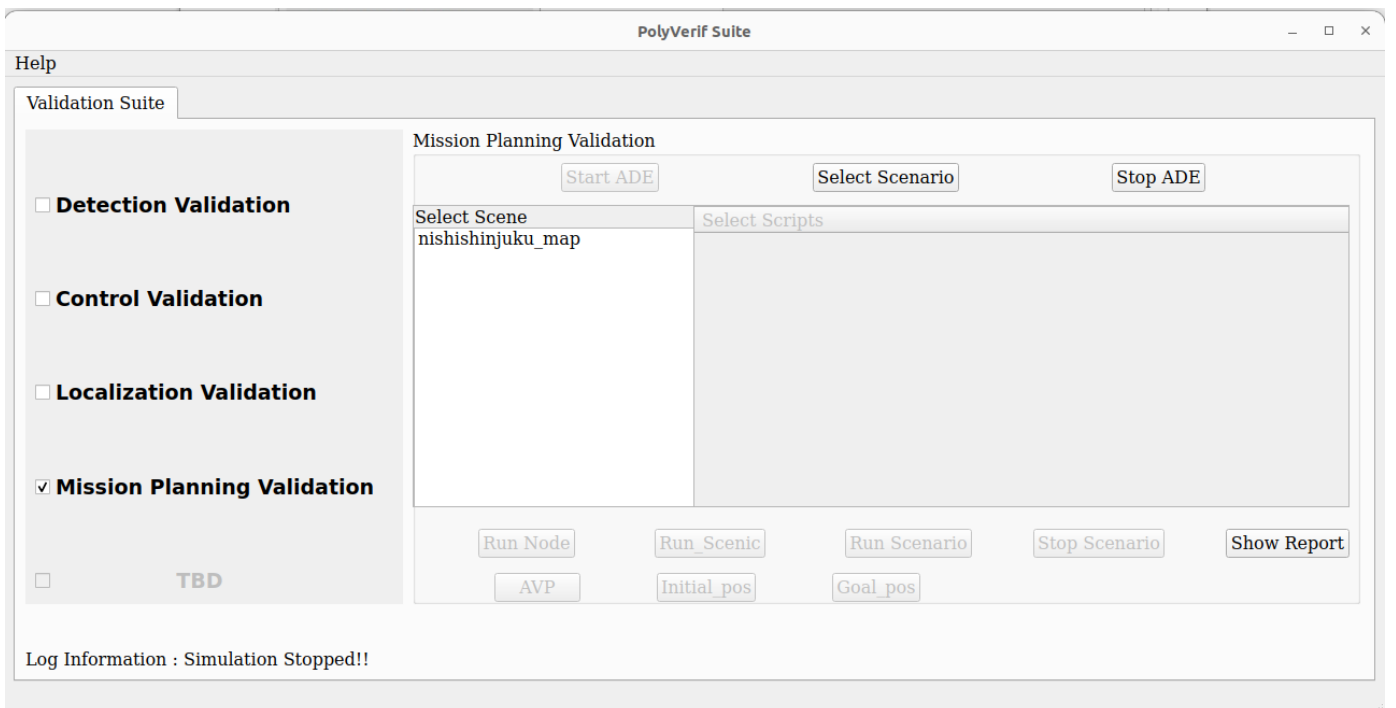


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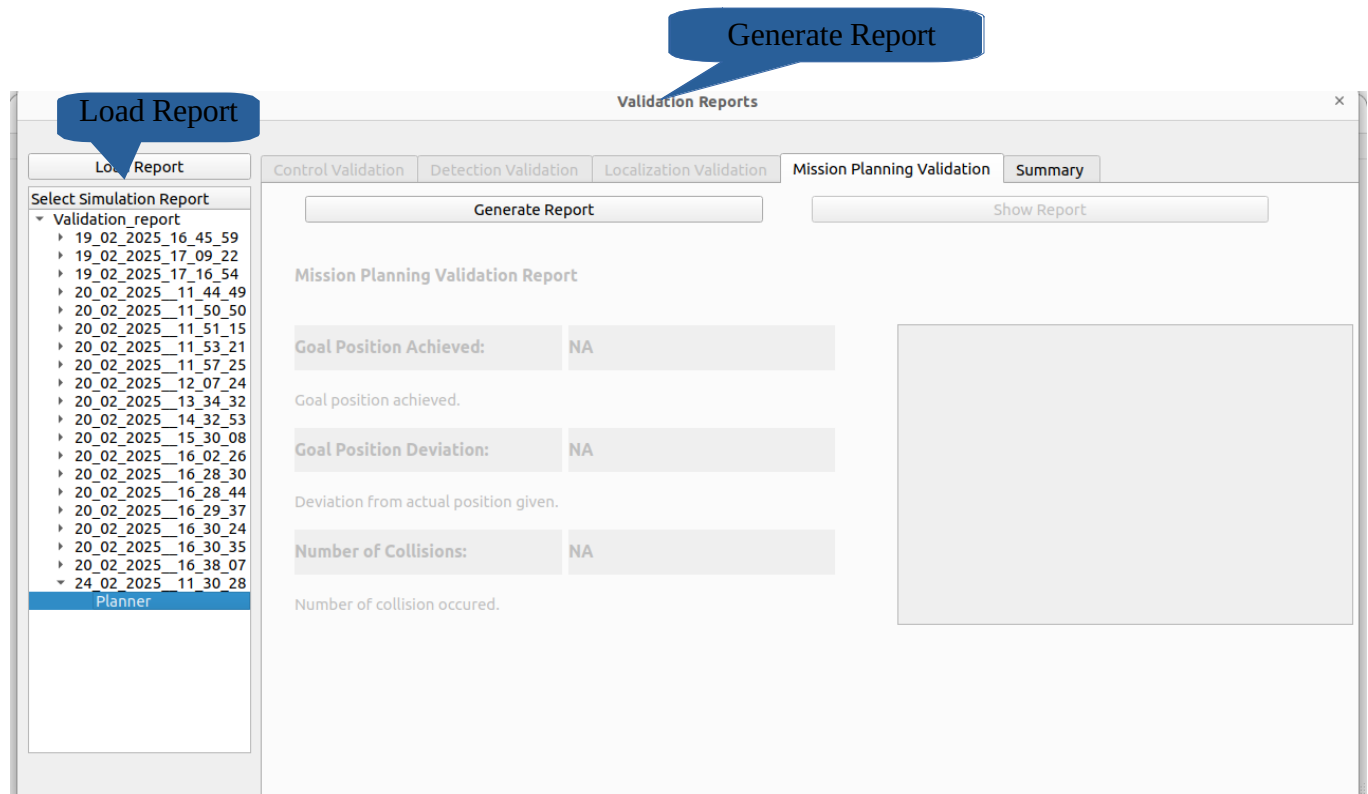


Access the detailed report through "Show Report."



2.Report Insights:

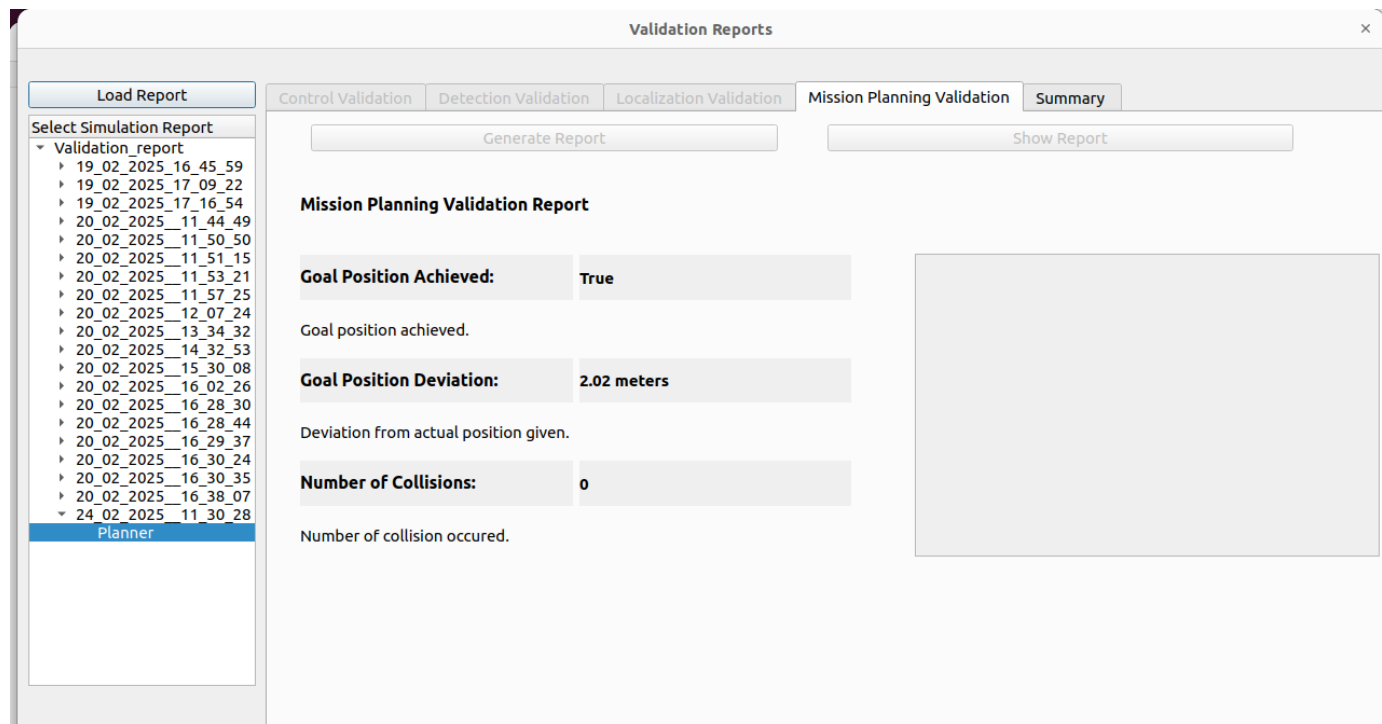
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3.Report Management:

Explore simulation reports.



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